

NUCLEOTIDE METABOLISM

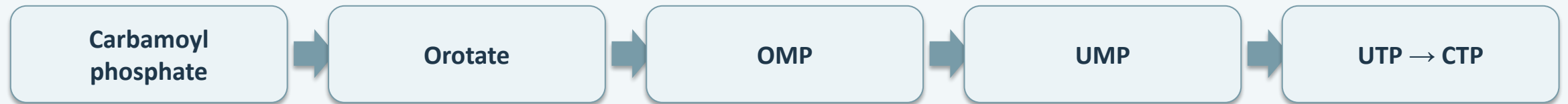
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Pyrimidine Nucleotide Biosynthesis & Metabolism



Pyrimidine Synthesis & Degradation

Nucleotide Metabolism



1. What are pyrimidines?

Pyrimidine bases

Cytosine (C), Uracil (U), and Thymine (T) are pyrimidines.

Where are they found?

- U is mainly in RNA
- T is mainly in DNA
- C is found in both

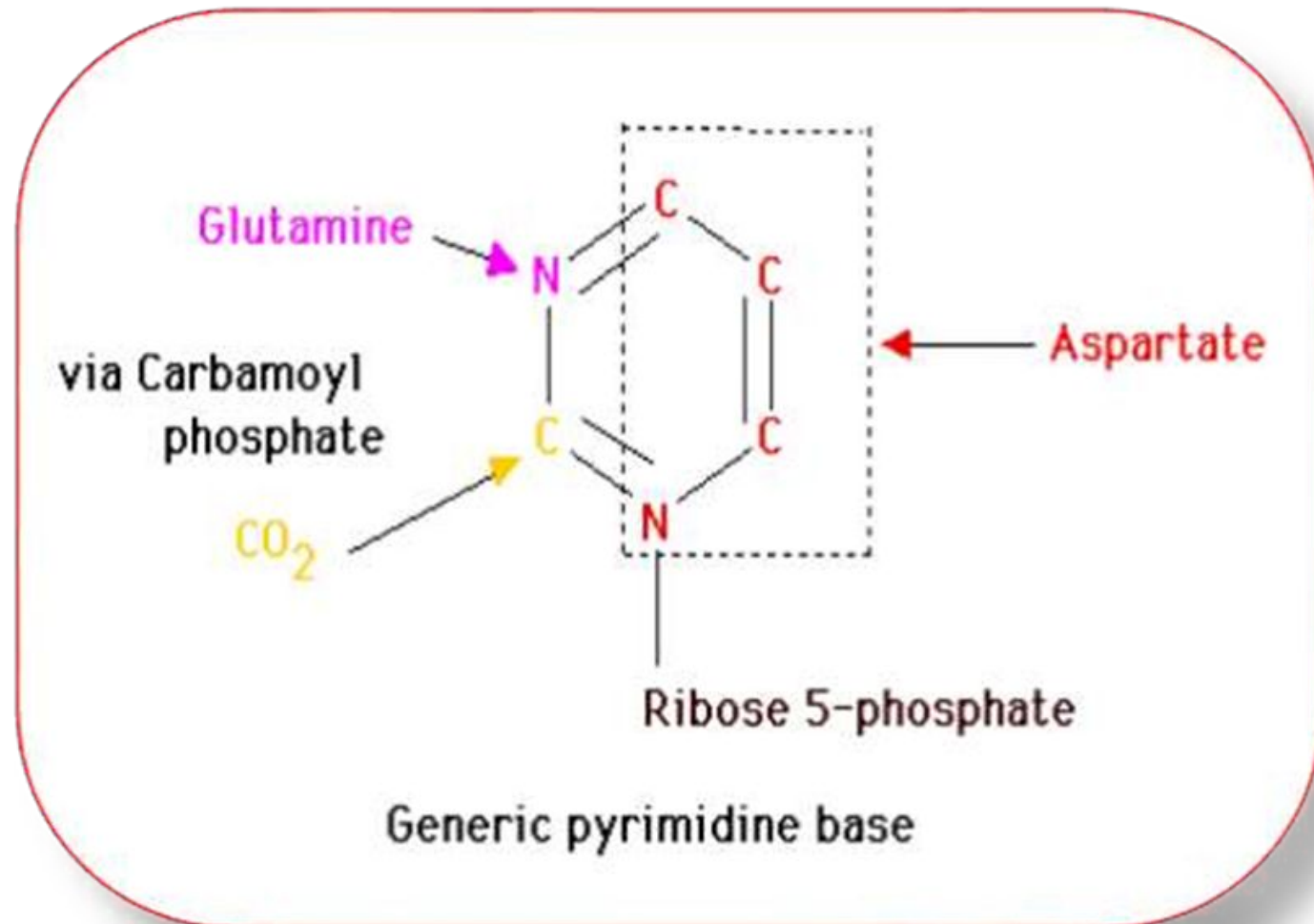
Key idea

In pyrimidine synthesis, the ring is built **FIRST** and then attached to ribose phosphate.

- This is different from purine synthesis, where the ring is built directly on ribose-5-phosphate.
- **PRPP (5-phosphoribosyl-1-pyrophosphate)** is an activated form of ribose-phosphate is used later.
- Main sources of pyrimidine-ring atoms: **glutamine**, **CO₂**, and **aspartate**.

So the sequence is:

Pyrimidine ring made → PRPP provides ribose-phosphate → complete pyrimidine nucleotide



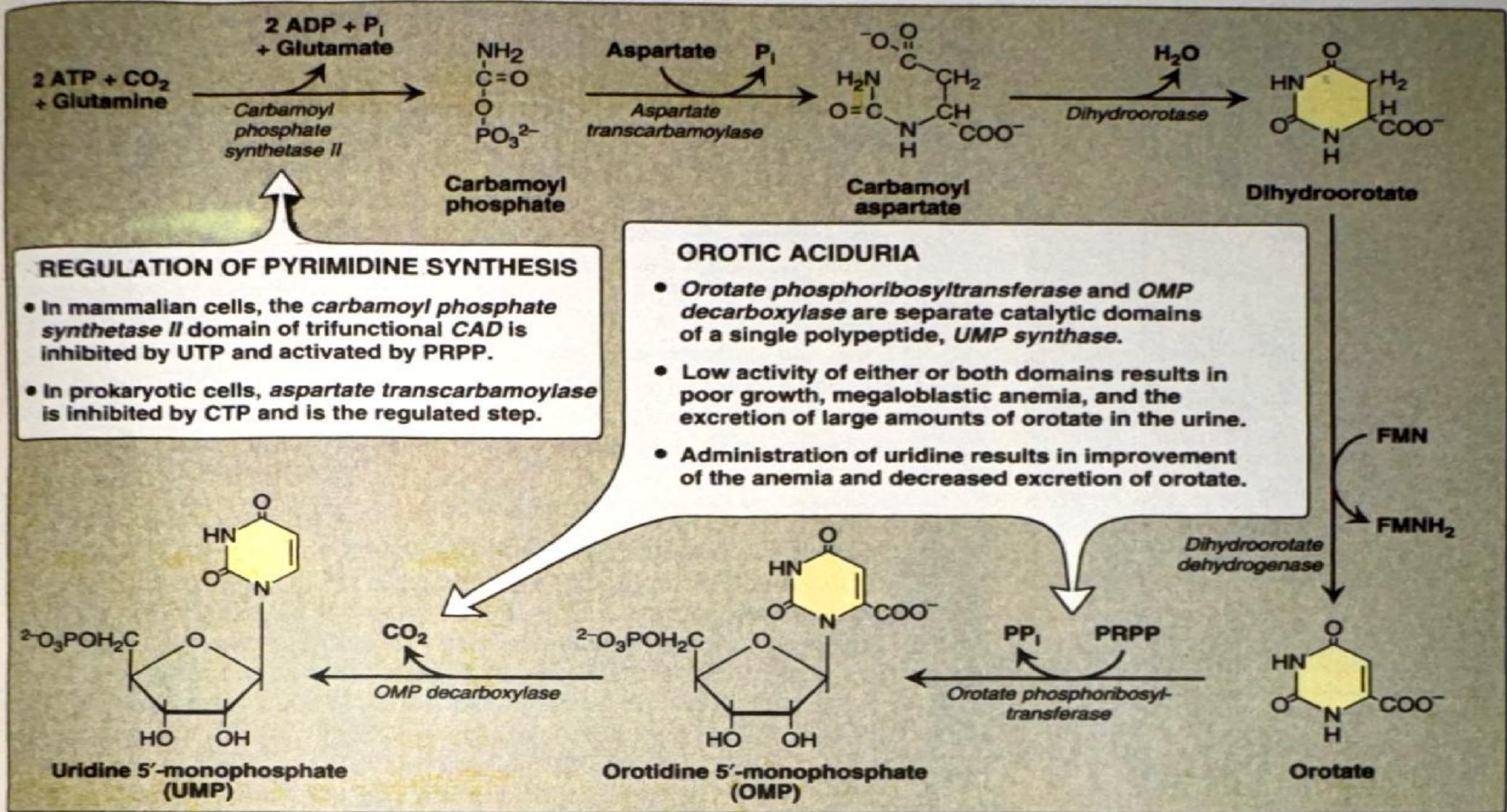


Figure 22.21

De novo pyrimidine synthesis. ADP = adenosine diphosphate; P_i = inorganic phosphate; FMN(H_2) = flavin mononucleotide; CTP = cytidine triphosphate; PRPP = 5-phosphoribosyl-1-pyrophosphate; PP_i = pyrophosphate.

2. De novo pyrimidine synthesis — overview



Regulated step

CPS II makes carbamoyl phosphate. It is **inhibited by UTP** and **activated by PRPP**.

Easy memory

Build the pyrimidine ring → attach it to PRPP → make UMP → make UTP/CTP.

3. Step 1 — Carbamoyl phosphate synthesis

Enzyme: CPS II

Carbamoyl phosphate synthetase II (CPS II) is in the cytosol and is part of the multifunctional CAD protein.

Reaction

$2 \text{ ATP} + \text{CO}_2 + \text{glutamine} \rightarrow \text{carbamoyl phosphate} + \text{products}$

- **Glutamine** supplies the **nitrogen**.
- This is the **regulated/rate-controlling step** in mammalian pyrimidine synthesis.
- **UTP** inhibits **CPS II** (feedback inhibition).
- **PRPP** activates **CPS II**.

4. CPS I vs CPS II — don't confuse them

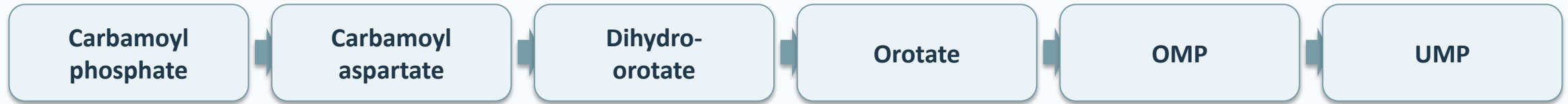
CPS I

- Location: **mitochondria**
- Pathway: **urea cycle**
- Nitrogen source: **ammonia**
- Activator: **N-acetylglutamate**
- Makes carbamoyl phosphate for the urea cycle

CPS II

- Location: **cytosol**
- Pathway: **pyrimidine synthesis**
- Nitrogen source: **amide group of glutamine**
- Activator: **PRPP**
- Inhibitor: **UTP**
- Part of **CAD protein**

5. Orotate → UMP



Key enzymes

Aspartate transcarbamoylase (ATCase) → dihydroorotase → dihydroorotate dehydrogenase → UMP synthase.

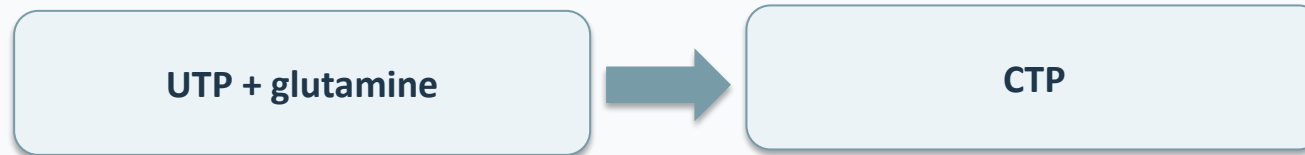
Important location

Most enzymes are **cytosolic**. **Dihydroorotate dehydrogenase** is associated with the **inner mitochondrial membrane**.

UMP synthase

Two activities: orotate phosphoribosyltransferase + OMP decarboxylase.

6. UMP is the starting point for other pyrimidine nucleotides



CTP synthase

Converts $\text{UTP} \rightarrow \text{CTP}$ using glutamine as the nitrogen source and ATP for energy.

For DNA

$\text{UDP} \rightarrow \text{dUDP} \rightarrow \text{dUTP} \rightarrow \text{dUMP}$. **dUMP** is the precursor for **dTMP(deoxythymidine monophosphate)**.

Why dUTP is controlled

dUTP is rapidly broken down to dUMP, helping prevent uracil from being inserted into DNA.

CTP Synthesis

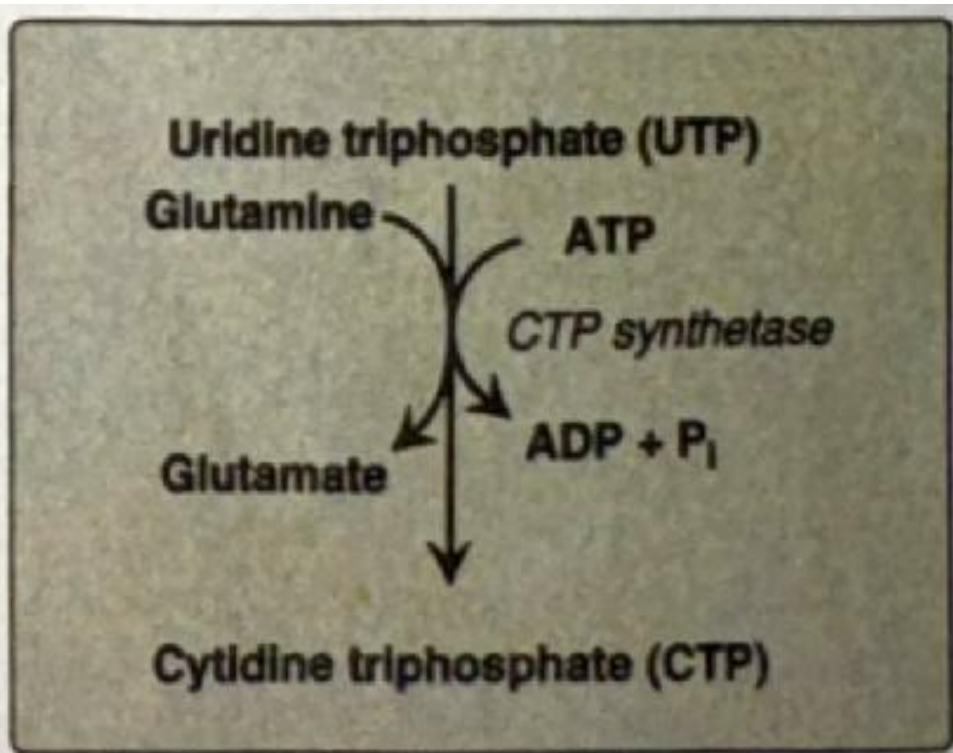
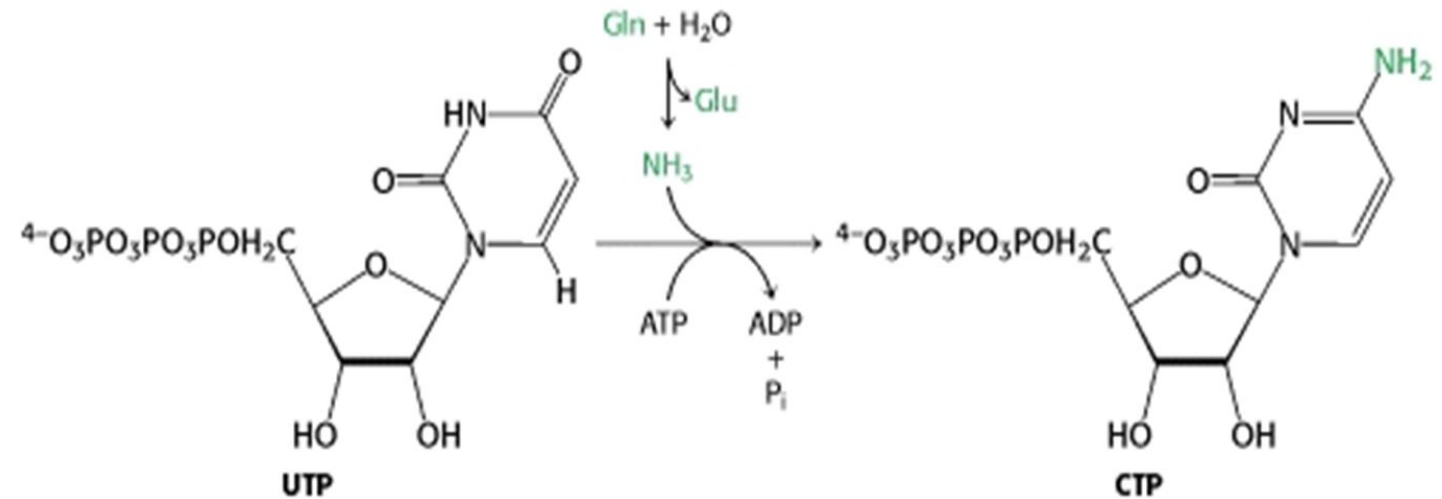
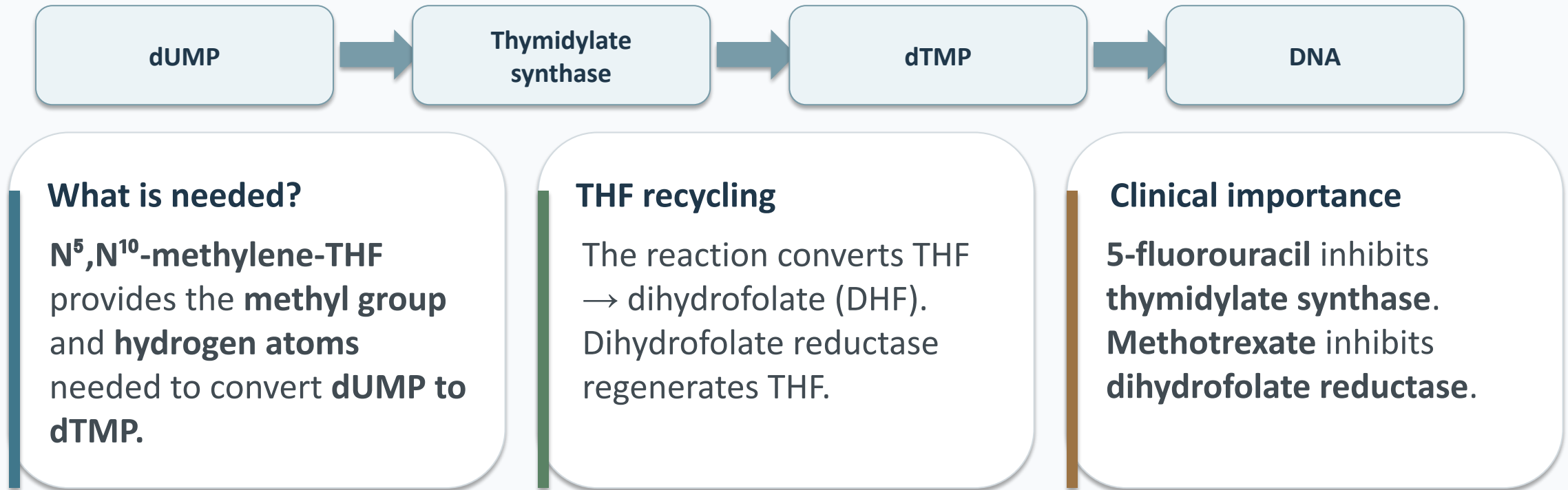


Figure 22.22

Synthesis of CTP from UTP. [Note: CTP, required for RNA synthesis, is converted to dCTP for DNA synthesis.] ADP = adenosine diphosphate; P_i = inorganic phosphate.



7. dTMP synthesis — making thymidine for DNA



- Blocking this pathway decreases DNA synthesis and slows rapidly dividing cells.
- That is why these drugs can be used as **anticancer agents**.

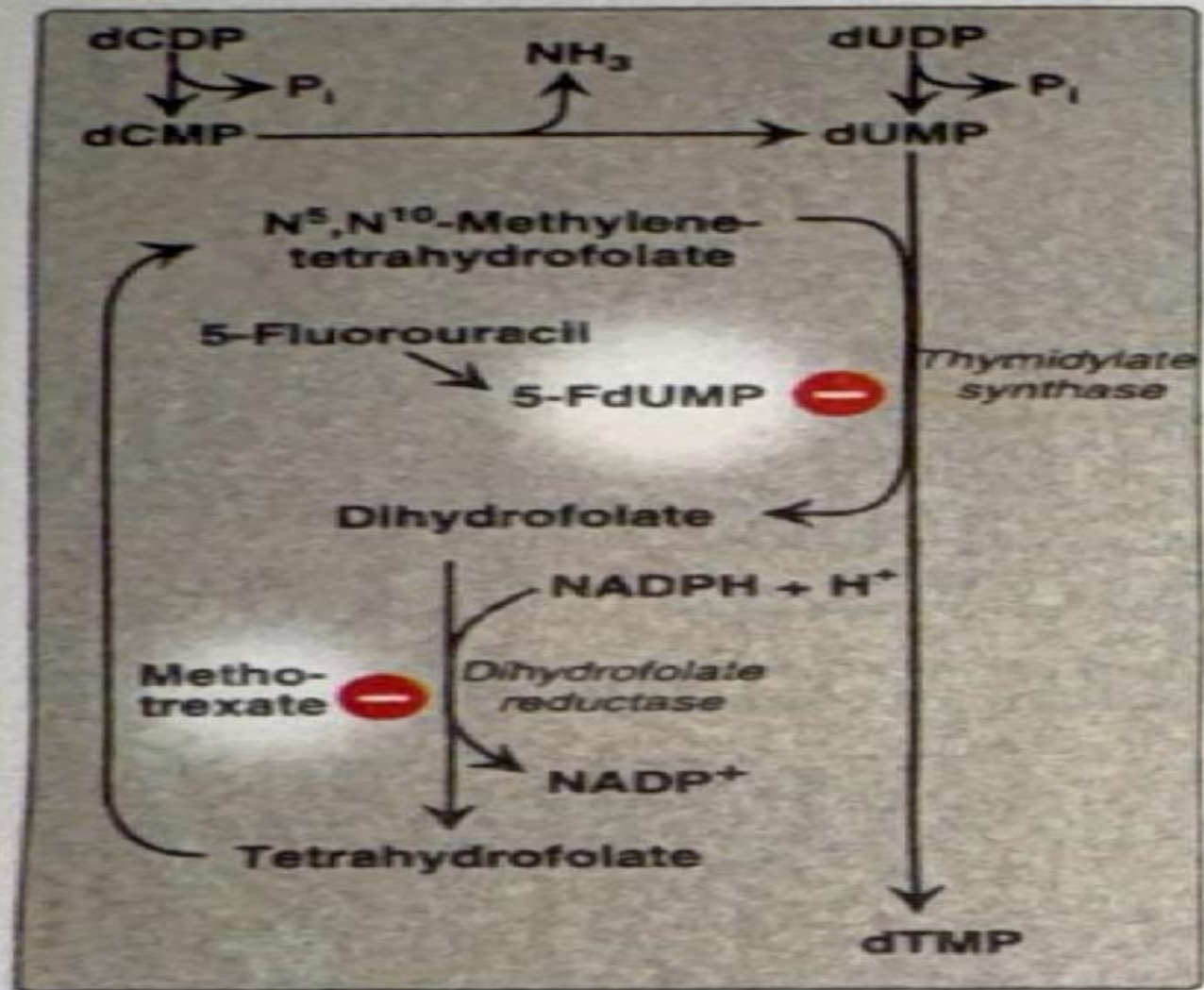


Figure 22.23

Synthesis of dTMP from dUMP, illustrating sites of action of antineoplastic drugs.

8. Orotic aciduria — an important disorder

What goes wrong?

UMP synthase has two catalytic activities.
Low activity of either one causes **orotic acid** to build up and appear in **urine**.

Main features

- Poor growth / growth failure
- Megaloblastic anemia
- Large amounts of orotate in urine

Treatment

Uridine administration can improve the anemia and reduce orotic acid excretion.

Memory point

Orotate ↑ + megaloblastic anemia + poor growth → think orotic aciduria.

9. Pyrimidine degradation and salvage

Degradation

Unlike purines, the pyrimidine ring is opened and broken down into highly soluble products.

Main products

- **β -alanine** — from degradation of **CMP** and **UMP**
- **β -aminoisobutyrate** — from **thymine/TMP** degradation
- **Ammonia** and **CO₂** are also produced.

Salvage

Pyrimidine bases can be reused by converting them to nucleosides and then phosphorylating them to nucleotides.

Clinical note

Pyrimidine salvage is less clinically important than purine salvage because pyrimidine breakdown products are very soluble.

10. Drugs that target nucleotide metabolism

5-Fluorouracil (5-FU)

Converted to 5-FdUMP, which inhibits thymidylate synthase $\rightarrow \downarrow$ dTMP $\rightarrow \downarrow$ DNA synthesis.

Methotrexate

Inhibits dihydrofolate reductase $\rightarrow \downarrow$ THF $\rightarrow \downarrow$ thymidine and purine synthesis $\rightarrow \downarrow$ DNA synthesis.

Why useful in cancer

Cancer cells divide rapidly, so blocking nucleotide production can slow their growth.

Acyclovir

A purine analog used against **herpes viruses**. It inhibits **viral DNA polymerase**.

AZT(Azidothymidine)

A pyrimidine analog used against **HIV**. It inhibits **viral DNA polymerase**.

Big picture

Nucleotide metabolism is an important drug target because DNA/RNA synthesis is essential for cell and viral replication.

11. High-yield summary

- Pyrimidine ring is built first, then attached to PRPP.
- CPS II is the regulated step of mammalian de novo pyrimidine synthesis.
- CPS II is inhibited by UTP and activated by PRPP.
- Pathway: carbamoyl phosphate $\rightarrow$ carbamoyl aspartate $\rightarrow$ dihydroorotate $\rightarrow$ orotate $\rightarrow$ OMP $\rightarrow$ UMP.
- UMP $\rightarrow$ UDP $\rightarrow$ UTP; UTP can be converted to CTP.
- dUMP $\rightarrow$ dTMP by thymidylate synthase; this step needs methylene-THF.
- 5-FU inhibits thymidylate synthase; methotrexate inhibits DHFR.
- Orotic aciduria is caused by UMP synthase deficiency and can be treated with uridine.
- Pyrimidines degrade to soluble products such as β -alanine and β -aminoisobutyrate.